

BEE 4750 Homework 1: Introduction to Using Julia

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Due Date

Thursday, 9/11/25, 9:00pm

Overview

Instructions

- Problem 1 consist of a series of code snippets for you to interpret and debug. You will be asked to identify relevant error(s) and fix the code.
- Problem 2 asks you to write code to implement several simple functions using more general programming and some which are unique to Julia (which might mean that you need to look at the documentation or use Julia's help function, which is accessed with `?`, e.g. `?sum` for help with the `sum()` function) or to analyze Julia syntax.
- Problem 3 asks you to convert a verbal description of a wastewater treatment system into a Julia function, and then to use that function to explore the impact of different wastewater allocation strategies.

Load Environment

The following code loads the environment and makes sure all needed packages are installed. This should be at the start of most Julia scripts.

```
import Pkg
Pkg.activate(@__DIR__)
Pkg.instantiate()
```

Activating project at `~/Desktop/Cornell\_Courses/FA\_25/BEE\_4750/bee4750-hw1-x1748`

Standard Julia practice is to load all needed packages at the top of a file. If you need to load any additional packages in any assignments beyond those which are loaded by default, feel free to add a `using` statement, though [you may need to install the package](#).

```
using Random
using Plots
using GraphRecipes
using LaTeXStrings
using Distributions
```

```
# this sets a random seed, which ensures reproducibility of random number generation. You should
Random.seed!(1)
```

```
TaskLocalRNG()
```

Problems (Total: 30 Points)

Problem 1 (12 points)

The following subproblems all involve code snippets that require debugging. You are encouraged to use online resources (*e.g.* Julia documentation and forums, Stack Overflow, Reddit, etc) to help find diagnose error messages and find solutions, but make sure that you clearly document which resources you used and how you used them.

Problem 1.1

You've been tasked with writing code to identify the minimum value in an array. You cannot use a predefined function. Your colleague suggested the function below, but it does not return the minimum value.

```
function minimum(array)
    # initialize the minimum value counter
    min_value = 0
    # update minimum values
    for i in 1:length(array)
        if array[i] < min_value
            min_value = array[i]
        end
    end
end
```

```

    end
    # return found minimum
    return min_value
end

array_values = [89, 90, 95, 100, 100, 78, 99, 98, 100, 95]
@show minimum(array_values);

```

```
minimum(array_values) = 0
```

What is the logical flaw in the code? Fix it and show that your fixed code solves the problem.

Writeup

The problem of this code is that the `min_value` is reset to 0 every time at the beginning of the function. When applying the `minimum` function to every item in the `array_values` list, they are always being compared to 0, and nothing was being saved, so the returned value is also 0. The code I changed set the minimum value to start at the first position of the list, thus allowing every following item since position 2 able to get compared to the previous value, and if the new value is smaller than the current minimum value, it will become the new value. So, in this case, the smallest integer in the list is 78.

```

# Fixed code
function minimum(array)
    # initialize the minimum value counter
    min_value = array[1]
    # update minimum values
    for i in 2:length(array)
        if array[i] < min_value
            min_value = array[i]
        end
    end
    # return found minimum
    return min_value
end

array_values = [89, 90, 95, 100, 100, 78, 99, 98, 100, 95]
@show minimum(array_values);

```

```
minimum(array_values) = 78
```

Problem 1.2

Your team is trying to compute the average grade for your class, but the following code produces an `UndefVarError`.

```
# enter student grade vector
student_grades = [89, 90, 95, 100, 100, 78, 99, 98, 100, 95]
# compute class average
function class_average(grades)
    average_grade = mean(student_grades)
    return average_grade
end

@show average_grade;
```

UndefVarError: UndefVarError: `average\_grade` not defined in `Main`

Suggestion: check for spelling errors or missing imports.

UndefVarError: `average\_grade` not defined in `Main`

Suggestion: check for spelling errors or missing imports.

Stacktrace:

[1] macro expansion

@ show.jl:1232 [inlined]

[2] top-level scope

@ ~/Desktop/Cornell_Courses/FA_25/BEE_4750/bee4750-hw1-xl748/jl_notebook_cell_df34fa98e69

What does this `UndefVarError` mean? Why does this code produce one? Fix the error and solve the problem.

Writeup

The `UndefVarError` is resulted from an unfound or undefined variable name, as signified by the error text. The error text mentions about how `average_age` is not found, which is because of how it is defined in the function. Because `average_age` equals the mean of `student_grades`, and the latter does not exist inside the function `class_average`, `average_age` is consequently

also not existing. My fixed code changes `student_grades` to be `grades`, which is the defined input variable for the function `class_average`. The calculated average for this list is 94.4.

```
# Fixed code
# enter student grade vector
student_grades = [89, 90, 95, 100, 100, 78, 99, 98, 100, 95]
# compute class average
function class_average(grades)
    average_grade = mean(grades)
    return average_grade
end

@show class_average(student_grades);
```

```
class_average(student_grades) = 94.4
```

Problem 1.3

Your team wants to know the expected payout of an old Italian dice game called *passadieci* (which was analyzed by Galileo as one of the first examples of a rigorous study of probability). The goal of *passadieci* is to get at least an 11 from rolling three fair, six-sided dice. Your strategy is to compute the average wins from 1,000 trials, but the code you've written below produces a `MethodError`.

```
# function to simulate a passedieci roll: 3 6-sided dice
function passedieci()
    # this rand() call samples 3 values from the vector [1, 6]
    roll = rand(1:6, 3)
    return roll
end
# set number of trials and initialize outcome vector
n_trials = 1_000
outcomes = zero(n_trials)
# simulate number of passedieci rolls and count wins
for i = 1:n_trials
    outcomes[i] = (sum(passadieci()) > 11)
end
win_prob = sum(outcomes) / n_trials # compute average number of wins
@show win_prob;
```

```
MethodError: MethodError: no method matching setindex!{::Int64, ::Bool, ::Int64}
```

The function `setindex!` exists, but no method is defined for this combination of argument types.
MethodError: no method matching setindex!(::Int64, ::Bool, ::Int64)

The function `setindex!` exists, but no method is defined for this combination of argument types.

Stacktrace:

[1] top-level scope

@ ~/Desktop/Cornell_Courses/FA_25/BEE_4750/bee4750-hw1-xl748/jl_notebook_cell_df34fa98e69

What does this `MethodError` mean? Why does this code produce one? Fix the error and solve the problem.

Writeup

I searched up and understood the `zero` function in Julia is used to return the additive identity of the input, which in this case, `outcomes` will turn out to be a single value which is 0. Since we want a list, this will not work. My fixed code changed it to use `zeros`, which creates a list of length 1000 and full of 0 as the integer type. Then `outcomes` can be updated in the loop each time. The final probability for winning fluctuates mainly from 35% to 40%.

```
# Fixed code
# function to simulate a passadieci roll: 3 6-sided dice
function passadieci()
    # this rand() call samples 3 values from the vector [1, 6]
    roll = rand(1:6, 3)
    return roll
end
# set number of trials and initialize outcome vector
n_trials = 1_000
outcomes = zeros{Int, n_trials}
# simulate number of passadieci rolls and count wins
for i = 1:n_trials
    outcomes[i] = (sum(passadieci()) > 11)
end
win_prob = sum(outcomes) / n_trials # compute average number of wins
@show win_prob;
```

win_prob = 0.382

Problem 1.4

You're interested in writing some code to remove the mean of a vector from all of its components. You've written the following code and tried to test it on a random vector, but your code returns a `MethodError`.

```
# function to remove mean from a vector
function remove_mean(vect)
    # function to compute the mean
    function compute_mean(vect)
        element_sum = 0 # initialize sum
        # compute mean and return
        for v in vect
            element_sum += v
        end
        return element_sum / length(vect)
    end

    m = compute_mean(vect) # compute mean
    # return demeaned vector
    return vect - m
end

random_vect = rand(1_000)
@show remove_mean(random_vect)
```

MethodError: MethodError: no method matching `- (::Vector{Float64}, ::Float64)`
For element-wise subtraction, use broadcasting with dot syntax: `array .- scalar`
The function ``-`` exists, but no method is defined for this combination of argument types.

Closest candidates are:

```
-(!Matched::ChainRulesCore.NoTangent, ::Any)
 @ ChainRulesCore ~/.julia/packages/ChainRulesCore/XAgYn/src/tangent_arithmetic.jl:61
- (::Any, !Matched::ChainRulesCore.NotImplemented)
 @ ChainRulesCore ~/.julia/packages/ChainRulesCore/XAgYn/src/tangent_arithmetic.jl:50
-(!Matched::ChainRulesCore.ZeroTangent, ::Any)
 @ ChainRulesCore ~/.julia/packages/ChainRulesCore/XAgYn/src/tangent_arithmetic.jl:101
...
```

MethodError: no method matching `- (::Vector{Float64}, ::Float64)`

For element-wise subtraction, use broadcasting with dot syntax: `array .- scalar`

The function `` exists, but no method is defined for this combination of argument types.

Closest candidates are:

```
-(!Matched::ChainRulesCore.NoTangent, ::Any)
  @ ChainRulesCore ~/.julia/packages/ChainRulesCore/XAgYn/src/tangent_arithmetic.jl:61
- (::Any, !Matched::ChainRulesCore.NotImplemented)
  @ ChainRulesCore ~/.julia/packages/ChainRulesCore/XAgYn/src/tangent_arithmetic.jl:50
-(!Matched::ChainRulesCore.ZeroTangent, ::Any)
  @ ChainRulesCore ~/.julia/packages/ChainRulesCore/XAgYn/src/tangent_arithmetic.jl:101
...
```

Stacktrace:

```
[1] remove_mean(vect::Vector{Float64})
  @ Main ~/.Desktop/Cornell_Courses/FA_25/BEE_4750/bee4750-hw1-x1748/jl_notebook_cell_df34fa98e69

[2] macro expansion
  @ show.jl:1232 [inlined]

[3] top-level scope
  @ ~/.Desktop/Cornell_Courses/FA_25/BEE_4750/bee4750-hw1-x1748/jl_notebook_cell_df34fa98e69
```

What does this `MethodError` mean? Why does this code produce one? Fix the error and solve the problem.

Writeup

The error happens when the vector `vect` is trying to subtract a single value `m`. In order to operate this constant subtracting process for every item of the vector, Julia requires a `.` in front of the operator, which is what happened in my fixed code, where I have `vect .- m`. The output is the updated vector with 1,000 components, with each component subtracted the mean value already.

```
# Fixed code
# function to remove mean from a vector
function remove_mean(vect)
    # fucntion to compute the mean
    function compute_mean(vect)
        element_sum = 0 # initialize sum
        # compute mean and return
        for v in vect
            element_sum += v
        end
        return element_sum / length(vect)
    end

    m = compute_mean(vect) # compute mean
    # return demeaned vector
    return vect .- m
end

random_vect = rand(1_000)
@show remove_mean(random_vect)
```

```
remove_mean(random_vect) = [-0.402858193077954, 0.17163216613218168, -0.18169592947668356, 0
```

```
1000-element Vector{Float64}:
```

```
-0.402858193077954
 0.17163216613218168
-0.18169592947668356
 0.060514380686886415
-0.2591266252711959
-0.01737820236308285
 0.37319528099391985
 0.23362003830781186
-0.3090151132440434
-0.06824664473090836
```

```
-0.12715199682996414
0.3553833867930265
0.2658817795877152
0.3071997855512709
0.35638130936574974
0.15303872493357518
-0.10958007508724765
-0.11638803141199039
0.4389100002859272
```

Problem 2 (18 points)

Cheap Plastic Products, Inc. is operating a plant that produces $100\text{m}^3/\text{day}$ of wastewater that is discharged into Pristine Brook. The wastewater contains $1\text{kg}/\text{m}^3$ of YUK, a toxic substance. The US Environmental Protection Agency has imposed an effluent standard on the plant prohibiting discharge of more than $20\text{kg}/\text{day}$ of YUK into Pristine Brook.

Cheap Plastic Products has analyzed two methods for reducing its discharges of YUK. Method 1 is land disposal, which costs $X_1^2/20$ dollars per day, where X_1 is the amount of wastewater disposed of on the land (m^3/day). With this method, 20% of the YUK applied to the land will eventually drain into the stream (*i.e.*, 80% of the YUK is removed by the soil).

Method 2 is a chemical treatment procedure which costs $\$1.50$ per m^3 of wastewater treated. The chemical treatment has an efficiency of $e = 1 - 0.005X_2$, where X_2 is the quantity of wastewater (m^3/day) treated. For example, if $X_2 = 50\text{m}^3/\text{day}$, then $e = 1 - 0.005(50) = 0.75$, so that 75% of the YUK is removed.

Cheap Plastic Products is wondering how to allocate their wastewater between these three disposal and treatment methods (land disposal, chemical treatment, and direct disposal) to meet the effluent standard while keeping costs manageable.

The flow of wastewater through this treatment system is shown in Figure 1. Modify the edge labels (by editing the `edge_labels` dictionary in the code producing Figure 1) to show how the wastewater allocations result in the final YUK discharge into Pristine Brook. For the `edge_label` dictionary, the tuple (i, j) corresponds to the arrow going from node i to node j . The syntax for any entry is $(i, j) \Rightarrow \text{"label text"}$, and the label text can include mathematical notation if the string is prefaced with an `L`, as in `L"x_1"` will produce x_1 .

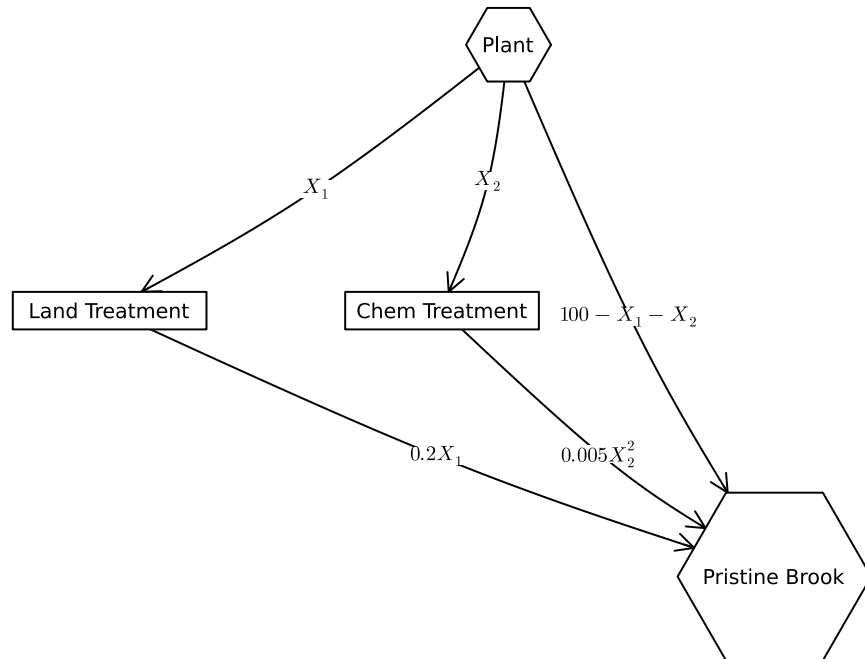
```
A = [0 1 1 1;
      0 0 0 1;
      0 0 0 1;
      0 0 0 0]
```

```

names = ["Plant", "Land Treatment", "Chem Treatment", "Pristine Brook"]
# modify this dictionary to add labels
edge_labels = Dict((1, 2) => L" $X_1$ ", (1, 3) => L" $X_2$ ", (1, 4) => L" $100 - X_1 - X_2$ ", (2, 4) => L" $0.2X_1$ ",
                  (3, 4) => L" $0.005X_2^2$ ")
shapes = [:hexagon, :rect, :rect, :hexagon]
xpos = [0, -1.5, -0.25, 1]
ypos = [1, 0, 0, -1]

p = graphplot(A, names = names, edgelabel = edge_labels, markersize = 0.15,
              markershapes = shapes, markercolor = :white, x = xpos, y = ypos)
display(p)

```



Problem 2.1

Formulate a mathematical model for the treatment cost and the amount of YUK that will be discharged into Pristine Brook based on the wastewater allocations. This is best done with some equations and supporting text explaining the derivation. Make sure you include, as additional equations in the model, any needed constraints on relevant values. You can find

some basics on writing mathematical equations using the LaTeX typesetting syntax [here](#), and a cheatsheet with LaTeX commands can be found on the course website's [Resources](#) page.

Based on the information and the graph given, we know that there are three ways of discharge into Pristine Brook. 1. Land disposal: The amount of wastewater going in from the plant is X_1 m³/day, and given the concentration of YUK is 1 kg/m³, we have a total of X_1 kg/day of YUK. In addition, given only 20% of the YUK applied here will go into the stream, we have $0.2X_1$ kg/day of YUK drained into Pristine Brook from method 1. The cost is given as $X_1^2/20$ dollars/day. 2. Chemical treatment: The amount of wastewater going in is X_2 m³/day, so the amount of YUK is X_2 kg/day. Given the efficiency equation $e = 1 - 0.005X_2$, with e being the percentage of YUK removed, the percentage of YUK drained into the stream is $1 - e = 0.005X_2$. We have thus $0.005X_2^2$ kg/day of YUK entering the stream from method 2. The cost is given as 1.5 dollars/m³ of wastewater, so it is $1.5X_2$ dollars/day. 3. Direct discharge: Given the total amount of wastewater being produced by the plant is 100 m³/day, the amount of direct wastewater discharge, without being treated, that goes straight into Pristine Brook is $100 - X_1 - X_2$ m³/day, which is $100 - X_1 - X_2$ kg/day of YUK discharged directly into Pristine Brook from the plant, at no cost of course.

To summarize, the mathematical model for the treatment cost and the amount of YUK that will be discharged into Pristine Brook based on the wastewater allocations can be assigned as:

$$\text{Minimize: } C(X_1, X_2) = \frac{X_1^2}{20} + 1.5X_2 \quad (\$/\text{day}) \quad (1)$$

$$\text{Subject to: } D(X_1, X_2) = -0.8X_1 - X_2 + 0.005X_2^2 + 100 \leq 20 \quad (\text{YUK discharge, kg/day}) \quad (2)$$

$$X_1 \geq 0, \quad X_2 \geq 0, \quad X_1 + X_2 \leq 100 \quad (3)$$

Problem 2.2

Implement your systems model as a Julia function which computes the resulting YUK discharge and cost for a particular treatment plan. You can return multiple values from a function with a [tuple](#), as in:

```
function multiple_return_values(x, y)
    return (x + y, x * y)
end

a, b = multiple_return_values(2, 5)
@show a;
@show b;
```

```
a = 7
b = 10
```

To evaluate the function over vectors of inputs, you can *broadcast* the function by adding a decimal `.` before the function arguments and accessing the resulting values by writing a *comprehension* to loop over the individual outputs in the vector:

```
x = [1, 2, 3, 4, 5]
y = [6, 7, 8, 9, 10]

output = multiple_return_values.(x, y)
a = [out[1] for out in output]
b = [out[2] for out in output]
@show a;
@show b;
```

```
a = [7, 9, 11, 13, 15]
b = [6, 14, 24, 36, 50]
```

Make sure you comment your code appropriately to make it clear what is going on and why.

```
# Function to compute YUK discharge and cost for a treatment plan
function treatment_plan(x1, x2)
    discharge = 0.2 * x1 + 0.005 * x2^2 + (100 - x1 - x2) # compute discharge
    cost = x1^2 / 20 + 1.5 * x2 # compute cost
    return discharge, cost # return a tuple of two values
end
```

```
treatment_plan (generic function with 1 method)
```

Writeup

My function `treatment_plan` takes two variables, x_1 and x_2 , which matches to the wastewater discharge into both methods of treatment. This function then calculates the YUK discharge based on the equation derived, as well as the costs. It then returns both variables.

Problem 2.3

Use your function to experiment with 1,000 different combinations of wastewater discharge and treatment. You can do this with either a grid search or by sampling from a [Dirichlet distribution](#) (a $\text{Dirichlet}(1, n)$ distribution will generate uniformly-weighted n -dimensional vectors whose components add up to 1; see the [Distributions.jl documentation](#) for how to sample from probability distributions in Julia). Plot the results of these experiments. Do any satisfy the YUK effluent standard (plot this as well as a dashed red line). What was the cost of solutions satisfying the standard? What can you say about the tradeoff between treatment cost and YUK discharge? You don't have to find an "optimal" solution to this problem, but what do you think would be needed to find a better solution?

```
using Distributions, Plots

# Function to compute YUK discharge and cost for a treatment plan
function treatment_plan(x1, x2)
    # Correct discharge calculation
    discharge = 0.2 * x1 + 0.005 * x2^2 + (100 - x1 - x2)
    cost = x1^2 / 20 + 1.5 * x2
    return discharge, cost
end

# Number of experiments
n = 1000

# Sample treatment allocations ensuring X1 + X2 = 100
X1 = zeros(n)
X2 = zeros(n)
for i in 1:n
    # Sample two proportions that sum to 1
    p = rand(Dirichlet(ones(2)))
    total = rand() * 100 # Random total allocation to treatment (100)
    X1[i] = p[1] * total
    X2[i] = p[2] * total
end

# Evaluate discharge and cost
discharge = zeros(n)
cost = zeros(n)
for i in 1:n
    discharge[i], cost[i] = treatment_plan(X1[i], X2[i])
end
```

```

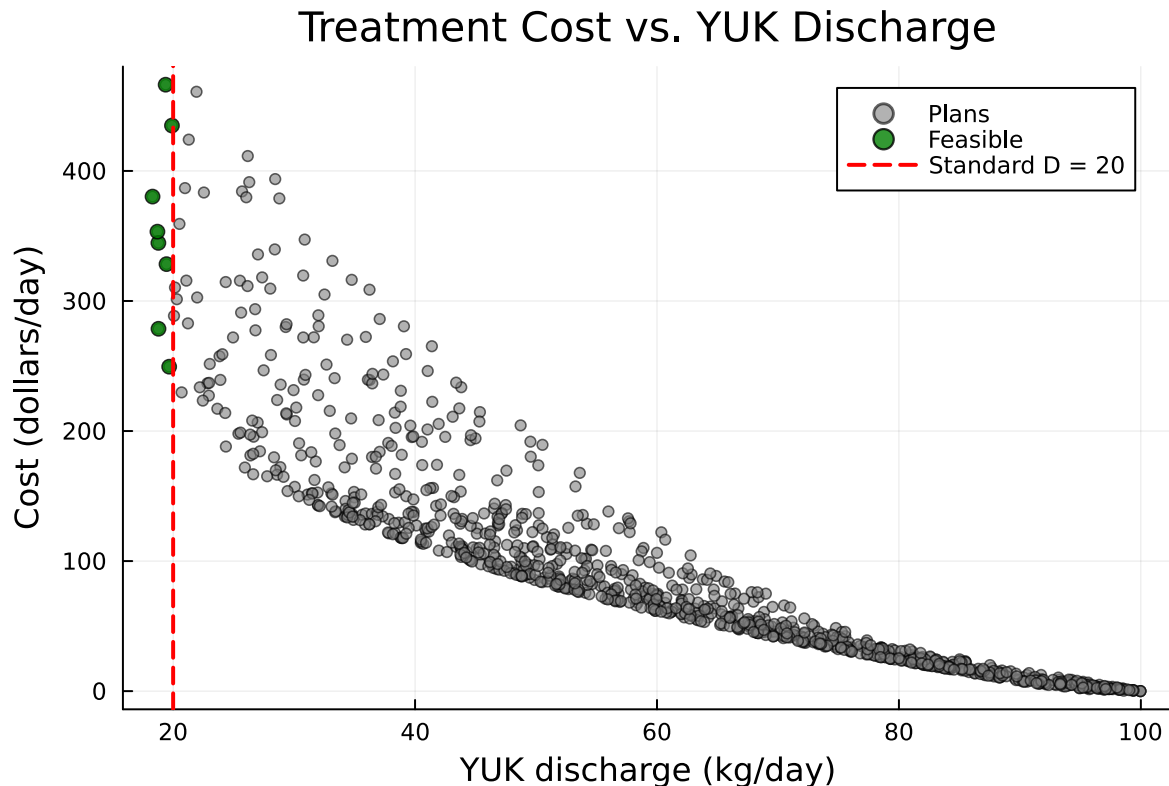
# Check feasibility: discharge 20 and allocations are valid
feasible = (discharge <= 20) & (X1 >= 0) & (X2 >= 0) & ((X1 + X2) <= 100)

# Plot all treatment plan possibilities
scatter(discharge, cost, ms = 3, alpha = 0.6, label = "Plans", color = :gray)

# Highlight feasible points
scatter!(discharge[feasible], cost[feasible],
         ms = 4, alpha = 0.8, color = :green, label = "Feasible")

# Draw YUK standard
vline!([20], linestyle = :dash, linewidth = 2, color = :red, label = "Standard D = 20")
xlabel!("YUK discharge (kg/day)")
ylabel!("Cost (dollars/day)")
title!("Treatment Cost vs. YUK Discharge")

```



Writeup

After simulating 1000 treatment allocations sampled from a Dirichlet distribution to ensure all wastewater is accounted for, I evaluated the resulting YUK discharge and cost for each plan. The results are plotted in the figure above, with the EPA's effluent standard of 20 kg/day shown as a red dashed line. Plans that meet this standard are highlighted in green.

For these feasible plans, the daily cost ranges approximately from \$250 to \$500. The overall cloud of points reveals a strong, non-linear tradeoff: achieving lower levels of YUK discharge comes at a significantly higher cost. The trend shows that initial reductions in discharge are very expensive, while the cost savings diminish as more pollution is allowed. This concave relationship is expected given the quadratic terms in the cost and discharge functions.

While this sampling approach helps visualize the feasible region and the cost-discharge relationship, it does not efficiently find the minimum cost. To identify the most cost-effective solution that precisely meets the discharge standard, a formal constrained optimization method would be necessary to minimize cost subject to the discharge constraint.

Problem 2.4

Find the strategies which minimize cost and YUK discharge (these will be different strategies) analytically and find the values of the objective metrics. Plot these values in the plot that you created for Problem 2.3. How do their values compare to the spread of values that you found in that problem? Would you select either of them (explain why or why not)?

```
# Cost-minimizing among feasible points
X1_costmin = 0.0
X2_costmin = 0.0
D_costmin, C_costmin = treatment_plan(X1_costmin, X2_costmin)

# YUK-minimizing among feasible points
min_YUK_idx = argmin(discharge[feasible]) # index of YUK discharge
X1_YUKmin = X1[feasible][min_YUK_idx]
X2_YUKmin = X2[feasible][min_YUK_idx]
D_YUKmin, C_YUKmin = treatment_plan(X1_YUKmin, X2_YUKmin)

println("Cost-min strategy: X1 = $(round(X1_costmin, digits = 2)), X2 = $(round(X2_costmin, digits = 2))",
        "Cost = $(round(C_costmin, digits = 2)), YUK = $(round(D_costmin, digits = 2))")
println("YUK-min strategy: X1 = $(round(X1_YUKmin, digits = 2)), X2 = $(round(X2_YUKmin, digits = 2))",
        "Cost = $(round(C_YUKmin, digits = 2)), YUK = $(round(D_YUKmin, digits = 2))")

# Plot for all plans
scatter(discharge, cost, ms = 3, alpha = 0.6, color = :gray, label = "All plans")
```



```

scatter!(discharge[feasible], cost[feasible], ms = 4, alpha = 0.8, color = :green, label = "Feasible")
vline!([20], linestyle = :dash, linewidth = 2, color = :red, label = "YUK standard D = 20")

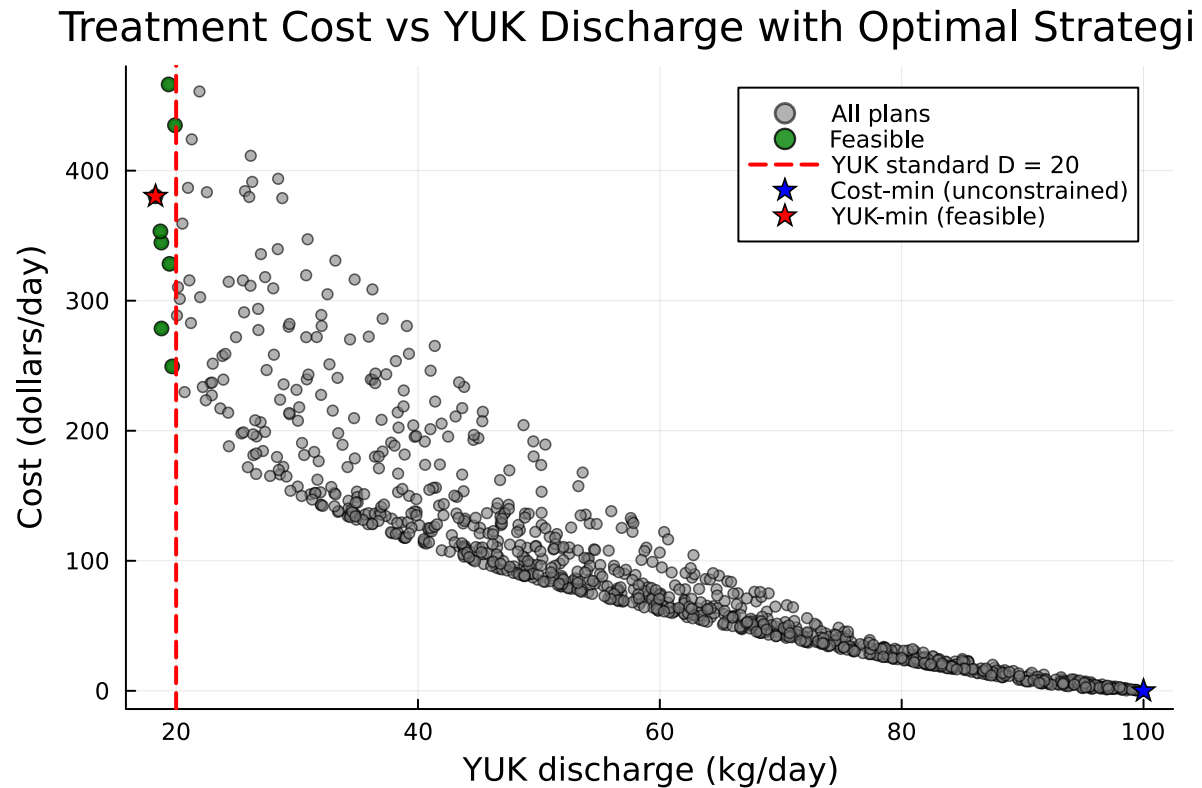
# Add optimized strategies
scatter!([D_costmin], [C_costmin], ms = 6, color = :blue, marker = :star5, label = "Cost-min (unconstrained)")
scatter!([D_YUKmin], [C_YUKmin], ms = 6, color = :red, marker = :star5, label = "YUK-min (feasible)")

xlabel!("YUK discharge (kg/day)")
ylabel!("Cost (dollars/day)")
title!("Treatment Cost vs YUK Discharge with Optimal Strategies")

```

Cost-min strategy: $X_1 = 0.0$, $X_2 = 0.0$, Cost = 0.0, YUK = 100.0

YUK-min strategy: $X_1 = 84.55$, $X_2 = 15.22$, Cost = 380.28, YUK = 18.3



Writeup

The goal is to find a strategy that minimizes cost while meeting the effluent standard of 20 kg/day of YUK. A strategy of direct discharge only ($X_1 = 0$, $X_2 = 0$) minimizes cost absolutely

to \$0/day but is infeasible, as it results in a discharge of 100 kg/day, far exceeding the permitted limit. Among the feasible strategies sampled (those with discharge ≤ 20 kg/day), the lowest discharge required allocations of $X_1 = 84.55\text{m}^3/\text{day}$ and $X_2 = 15.22\text{m}^3/\text{day}$, resulting in a discharge of 18.3 kg/day at a high cost of \$380.28/day.

Comparing these two extremes from the sample reveals a significant trade-off. The YUK-minimizing strategy is very expensive for only a marginal improvement in cleanliness compared to the standard. The scatter plot shows a cluster of points suggesting cost decreases rapidly as discharge increases from ~ 20 kg/day to ~ 40 kg/day. Since I believe neither of the two extreme sample points is likely the ideal operational choice, many of the other points offer a much more favorable balance; for example, achieving a discharge of ~ 30 kg/day for around \$150/day represents a substantial cost saving over the min-YUK strategy for a relatively moderate increase in pollution.

References

List any external references consulted, including classmates.

1. Used Julia Handbook for problem 2.3 for helping to understand the Dirichlet distribution. Distributions.jl. (2025, May 11). *Dirichlet*. In *Multivariate Distributions*. JuliaStats. Retrieved from <https://juliastats.org/Distributions.jl/stable/multivariate/#Distributions.Dirichlet>
2. Used ChatGPT for problems 1.3, 2.2, 2.4, mainly for searching syntax of Julia and helping understand how the coding language works, not for discussions and analysis of problems. OpenAI. (2025). ChatGPT (Feb 1 GPT-4 model) [Large language model]. <https://chat.openai.com>